Multi-Curves Framework

Advanced Financial Modeling

Andrea Carapelli

Università di Siena

2023

UNIVERSITÀ DI SIENA 1240

1 Introduction

2 Single Curve Context

3 Towards the Multiple Curves framework

4 Convexity Adjustments

Suggested References

Everything will be covered during the course and provided to you as slides, notes or Python notebooks.
If you want more, check the following:

- D. Brigo, F. Mercurio (2006) Interest Rate Models - Theory and Practice (2nd edition)
- L. Andersen, V. Piterbarg (2010) Interest Rate Models. Vol.1, Vol.2, Vol.3.

Building Blocks - Simple Rates

Recall that the simply compounded spot interest rate between t and T is denoted by $L(t,T)$ and represent the constant rate at which the amount $P(t,T)$ at time t is equivalent to one unit at time T.

$$L(t,T) = \frac{P(T,T) - P(t,T)}{\tau(t,T)P(t,T)}, \qquad P(T,T) = 1$$

Where $P(t,T)$ represent a zero-coupon bond price which is given by

$$P(t,T) = \frac{1}{1 + \tau(t,T)L(t,T)}$$

Building Blocks - Forwards Reminder

Starting from the definition of forward rate agreement (FRA), by absence of arbitrage we obtain the following formula of the value of the FRA at time t

$$\mathbf{FRA}(t, T, S, \tau, N, K) = N\big[P(t, S)\tau(T, S)K - P(t, T) + P(t, S)\big]$$

we can define the forward rate $F(t, T, S)$ with $t \leq T \leq S$ as the fixed rate that makes the contract fair at inception

$$F(t, T, S) = \frac{1}{\tau(T, S)}\left[\frac{P(t, T)}{P(t, S)} - 1\right] \tag{1}$$

The instantaneous forward interest rate $f(t, T)$ is a forward rate with maturity with $T + \Delta T$, where ΔT is small

$$f(t, T) = f(t, T, T + \Delta T) - \frac{\partial lnP(t, T)}{\partial T}$$

Building Blocks - Swaps Reminder

At time t the value of the floating leg is equal to the discounted expectation of future LIBOR fixing rates. For a single payment:

$$\mathbf{FL}(t, T_{k-1}, T_k) = \tau_k P(t, T_k) \mathbb{E}^{T_k}[L(T_{k-1}, T_k)]$$

Under the forward measure $\mathbb{Q}^{T_k}$ the expected value of future LIBOR rate is the forward rate observed in t, fixed in T_{k-1}, to be exchanged in T_k

$$\mathbf{FL}(t, T_{k-1}, T_k) = \tau_k P(t, T_k) F(t, T_{k-1}, T_k)$$

The floating leg net present value is obtained by summing the value of all flows

$$\mathbf{FL}(t, T_a, \ldots, T_b) = \sum_{k=a+1}^{b} \tau_k P(t, T_k) F(t, T_{k-1}, T_k)$$

Building Blocks - Swaps Reminder

Now consider the swap's fixed leg with K being the fixed rate paid on each payment date $T_c \ldots, T_d$., the net present value is equal to the discounted future cash flows

$$\mathbf{FIX}(t, T_c, \ldots, T_d) = K \sum_{j=c+1}^{d} \tau_j P(t, T_j)$$

Now we can exploit the price of a payer swap as the difference between the floating leg and the fixed leg

$$\mathbf{IRS}(t, T_a, \ldots, T_b, T_c, \ldots, T_d) = \sum_{k=a+1}^{b} \tau_k P(t, T_k) F(t, T_{k-1}, T_k) - K \sum_{j=c+1}^{d} \tau_j P(t, T_j)$$

Building Blocks - Swaps Reminder

We can define the swap rate S as the fair rate K that makes the value of the swap equal to zero at inception

$$S_{a,b,c,d}(t) = \frac{\sum_{k=a+1}^{b} \tau_k P(t, T_k) F(t, T_{k-1}, T_k)}{\sum_{j=c+1}^{d} \tau_j P(t, T_j)}$$

1 Introduction

2 Single Curve Context

3 Towards the Multiple Curves framework

4 Convexity Adjustments

Single Curve

At the foundation of interest rate trading there are two curves:

- discounting curve is used to discount the future cash-flows in order to obtain their present value
- forwarding (or estimation) curve is used to obtain the expected value of the forward rates at future times.

In a market context where

- Interbank credit/liquidity issues did not matter for pricing, basis swap spreads were negligible (and neglected).
- The collateral did not matter for pricing.

Libor rates were a good proxy for risk free rates abd was adopted both for discounting and forwarding purposes.

Single Curve

The rule of thumb was the following:

- Select one finite set of the most convenient (i.e. liquid) vanilla interest rate instruments traded in real time on the market, with increasing maturities.
- Interpolate/bootstrap a single yield curve for each currency using the selected instruments.
- Use the same curve to compute cash flows and discount factors.
- Hedge the resulting delta risk using the necessary hedge ratios of the same set of vanillas.

Single Curve Inconsistencies

The pre-crisis approach outlined above is no longer consistent, at least in this simple formulation, with the present market approach:

1. it does not take into account the market information carried by the basis swap spreads,
2. it does not take into account that the interest rate market is segmented into sub-areas corresponding to instruments with different underlying rate tenors, characterized, in principle, by different dynamics (like correlated processes, we'll see later in the course).
3. it does not include collateral agreements and funding rates associated with the bootstrapping instruments.

1 Introduction

2 Single Curve Context

3 Towards the Multiple Curves framework

4 Convexity Adjustments

Modern Market Practice

The post-crisis standard market practice for the construction of multiple interest rate yield curves can be summarized in the following procedure:

- select multiple separated sets of vanilla interest rate (typically with 1M, 3M, 6M, 12M tenors),
- for each currency, build multiple separated forwarding curves using the selected instruments,
- compute the relevant forward rates and the corresponding cash flows from the forwarding curve with the appropriate tenor,
- compute the relevant discount factors from the discounting curve with the appropriate funding characteristics.

Yield Curve Construction

The key point in the procedure above (both in single and multi curves) is much more a matter of art than of science [1]. The main criteria to use in judging a curve construction are:

- the yield curve should be able to price back the instruments which are used to construct it.
- implied forward rates must be continuous.
- the interpolation used should be as local as possible,
- The hedge should be local.

[1] See Ametrano-Bianchetti (2013) for further details.

Discounting

Nowadays the majority of interest rate derivatives are collateralized, for this reason the common curve used for discounting is the OIS curve (C_d). The OIS rate (in EU) is defined as the following geometric sum

$$r_{ois}(t,T) = \left[\prod_{i=t}^{T-\frac{1}{360}} \left(1 + \frac{r_{ON_i}}{360}\right) - 1 \right] \frac{360}{n}$$

where r_{ON_j} is the reference rate from T_i to $T_i + 1day$ and n is the number of days in the calculation period.

Discounting

Starting from the rates of this quoted pillars is possible to get all the corresponding discount factors, identified by the unitary zero-coupon bonds $P_d(t, T_i)$ associated with $r_{ois}(t, T_i)$

$$\begin{aligned} P_d(t, T_i) &= e^{-\int_t^{T_i} r_{ois}(s, s+1d)ds} \\ &= e^{-r_{ois}(t, T_i)(T_i - t)} \end{aligned}$$

and consequently the FRA rates

$$F_d(t, T_{i-1}, T_i) = \frac{1}{\tau(T_{i-1}, T_i)} \left(\frac{P_d(t, T_{i-1})}{P_d(t, T_i)} - 1 \right) \tag{2}$$

Discounting - Deposits

The first stripping instrument is a simple spot cash deposit with 1 day maturity, a zero coupon bond in which a counter-part lends a nominal amount (N) to another counter-part, which at maturity pays the notional amount plus the $r_{ois}(t, t+1)$ rate accrued over this period. The payoff is given by

$$\phi_{Deposit}(t+1) = N(1 + r_{ois}(t, t+1)\tau_{1day})$$

and its t-value is

$$\begin{aligned}\phi_{Deposit}(t) &= P_d(t, t+1)\mathbb{E}^{t+1}\left[\frac{\phi_{Deposit}(t+1)}{P_d(t+1, t+1)}\right] \\ &= NP_d(t, t+1)(1 + r_{ois}(t, t+1)\tau_{1day}) = N\end{aligned}$$

Following that

$$P_d(t, T_i) = \frac{1}{1 + r_{ois}(t, T_i)\tau_i}$$

Discounting - OIS Swaps

We can exploit the price of a OIS payer swap as the difference between the floating leg and the fixed leg

$$\mathbf{IRS}_{OIS}(t, T_a, \ldots, T_b, T_c, \ldots, T_d) = \sum_{k=a+1}^{b} \tau_k P_d(t, T_k) F_d(t, T_{k-1}, T_k) - K \sum_{j=c+1}^{d} \tau_j P_d(t, T_j)$$

Discounting - OIS Swaps

From here it's easy to derive the value of K which makes the contract fair in t, i.e. the fixed rate s.t. $\mathbf{IRS}_{OIS} = 0$, the so called SWAP rate, given by

$$S_{a,b,c,d}(t) = \frac{\sum_{k=a+1}^{b} \tau_k P_d(t, T_k) F(t, T_{k-1}, T_k)}{A_{c,d}(t)}$$

Where $A_{c,d}(t) = \sum_{j=c+1}^{d} \tau_j P_d(t, T_j)$ is called the *annuity factor*.

Forwarding

In constructing the OIS Discount Curve, both discounting factors and forward rates are compounded starting from the same curve (C_d), which means that we still worked in a single curve framework. Things are different when we deal with forward curves with different tenors (e.g. 1M, 3M, 6M, 12M):

- we will need the C_d curve we just build for discounting, which we have assumed to be the best proxy of the risk-free rate
- then we need an additional curve C_x for the forward rates with the desired tenor x

Forwarding

The basic forward curve is usually bootstrapped using a selection from the following market instruments:

- deposit contracts, covering the window from today up to 1Y,
- FRA contracts,covering the window from 1M up to 2Y,
- IRS contracts, covering the window from 2Y up to 60Y,

the selection is generally done according to the principle of maximum liquidity.

Forwarding - FRA

FRA contracts are nowadays, in most cases, regulated by a CSA, so if we assume continuous mark-to-market and collateral posting, then we could neglect counterparty default risk, and discount our cashflow using the risk free discount factor $P_d(t, T_n)$. Hence the t-value of the FRA (with tenor $6m$) will be given by

$$FRA_{6m} = P_d(t, T_n) \frac{k - F_{6m}(t, T_n, T_S)}{1 + \tau_{6m} F_{6m}(t, T_n, T_S)} \tau_{6m}$$

where the FRA rate, k, that makes the contract fair, is given by

$$k = F_{6m}(t, T_n, T_S) = \frac{1}{\tau_{6m}} \left(\frac{P_{6m}(t, T_n)}{P_{6m}(t, T_S)} - 1 \right)$$

Forwarding - FRA

remember that $P_{6m}(t, T_n)$ is the unitary zero-coupon bond associated with $r_{6m}(t, T_n)$

$$P_{6m}(t, T_n) = e^{-\int_t^{T_n} r_{6m}(s,s+6m)ds} = e^{r_{6m}(t,T_n)(T_n-t)}$$

Forwarding - SWAPS

In $6M$ SWAPs the semi-annual index is exchanged against an annual coupon, the two leg have different payment frequencies (6m vs 1y), so we have that:

$$
\begin{aligned}
SWAP_{6m}(t, T_n, T_S, k) &= FIX_{6m}(t, T_n, T_S, k) - FLT_{6m}(t, T_n, T_S) \\
&= k \sum_{j=n+1}^{S} P_d(t, T_j) - \sum_{j=n+1}^{S} P_d(t, T_j) \left(\frac{P_{6m}(t, T_{j-1})}{P_{6m}(t, T_j)} - 1 \right) \\
&= kA_{d,n,S}(t) - \sum_{j=n+1}^{S} P_d(t, T_j) \left(\frac{P_{6m}(t, T_{j-1})}{P_{6m}(t, T_j)} - 1 \right)
\end{aligned}
$$

Forwarding - SWAPS

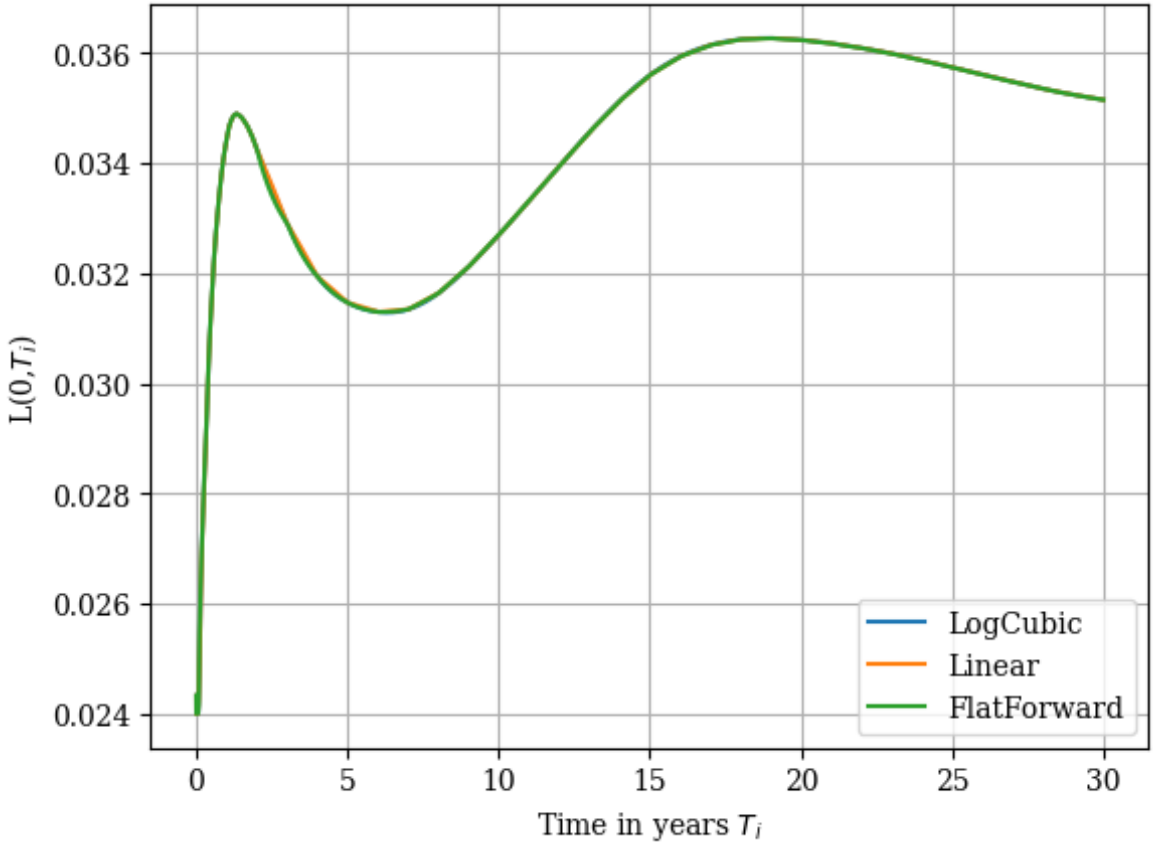
And the strike, k, that makes the SWAP fair will be

$$k = \frac{\sum_{j=n+1}^{S} P_d(t, T_j)\left(\frac{P_{6m}(t,T_{j-1})}{P_{6m}(t,T_j)} - 1\right)}{A_{d,n,S}(t)}$$

where $A_{d,n,S}(t)$ is the annuity factor computed from the discounting curve.

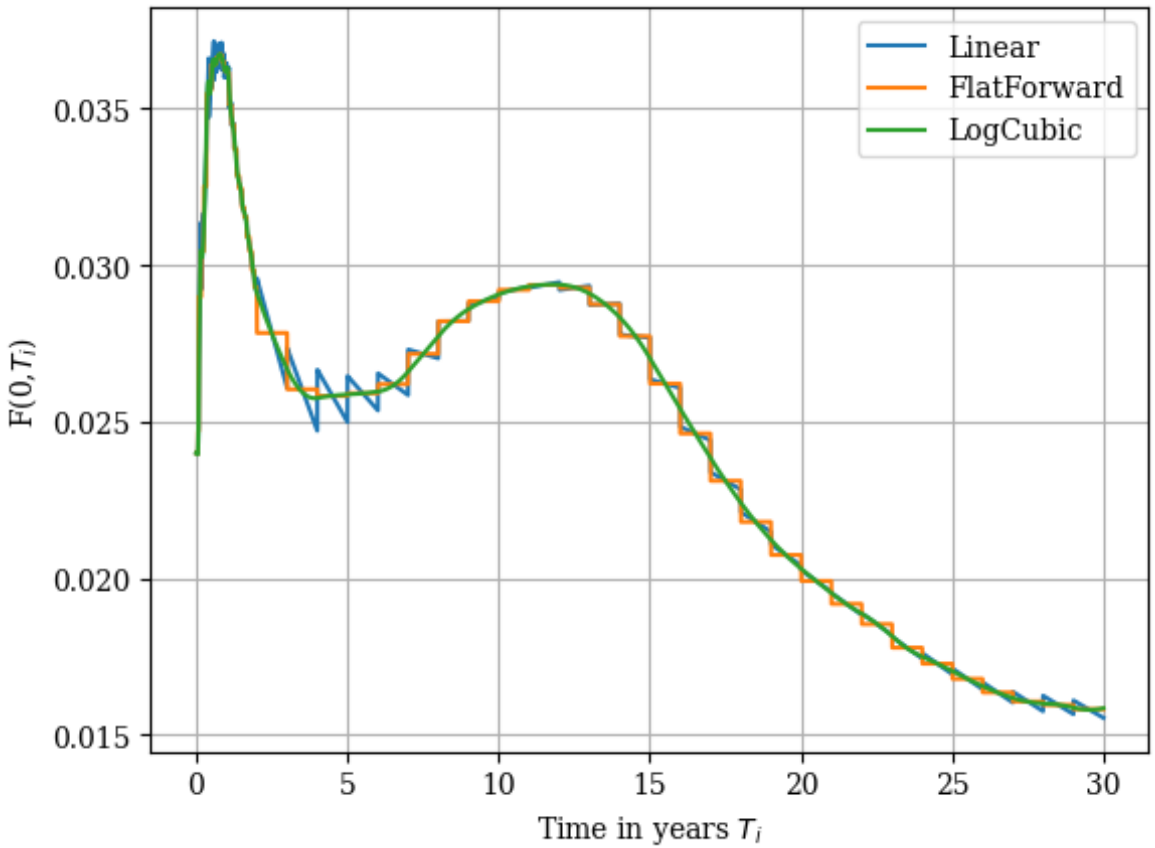
Choice of Interpolation

The choice of the interpolation function does not seems to be important for zero rates



Choice of Interpolation

But has an high importance in determining the smoothness of our forward rates structure.



1 Introduction

2 Single Curve Context

3 Towards the Multiple Curves framework

4 Convexity Adjustments

Gentle Introduction

Introduction

Let us consider a basic interest rate payoff function, which pays a percentage of a notional N, and the percentage paid will be determined by the Libor rate $L(T_{i-1}, T_i)$, at time T_i.

$$\begin{aligned} V(t_0) &= NM(t_0)\mathbb{E}^{\mathbb{Q}}\left[\frac{L(T_{i-1}, T_i)}{M(T_i)}\right] \\ &= NP(t_0, T_i)\mathbb{E}^{T_i}[L(T_{i-1}, T_i)] \end{aligned}$$

We know that under the forward measure $\mathbb{Q}^{T_i}$ the price of such a contract is given by:

$$V(t_0) = NP(t_0, T_i)F(t_0, T_{i-1}, T_i)$$

Different Numeraire?

Suppose now that we consider the same contract, however, the payment will take place at some earlier time $T_{i-1} < T_i$, the current value of the contract is then given by:

$$V(t_0) = NM(t_0)\mathbb{E}^{\mathbb{Q}}\left[\frac{L(T_{i-1}, T_i)}{M(T_{i-1})}\right]$$

When changing measures, to the T_{i-1} forward measure, we work with the following Radon-Nikodym derivative:

$$\left.\frac{d\mathbb{Q}^{T_{i-1}}}{d\mathbb{Q}}\right|_{\mathcal{F}(t_{i-1})} = \frac{P(T_{i-1}, T_{i-1})}{P(t_0, T_{i-1})}\frac{M(t_0)}{M(T_{i-1})}$$

so that

$$V(t_0) = NM(t_0)\mathbb{E}^{T_{i-1}}\left[\frac{P(t_0, T_{i-1})}{P(T_{i-1}, T_{i-1})}\frac{M(T_{i-1})}{M(t_0)}\frac{L(T_{i-1}, T_i)}{M(T_{i-1})}\right]$$

Different Measure!

therefore

$$V(t_0) = NP(t_0, T_{i-1})\mathbb{E}^{T_{i-1}}[L(T_{i-1}, T_i)] \tag{3}$$

Although the Libor rate $L(T_{i-1}, T_i)$ is a martingale under the T_i forward measure, it is however not a martingale under the T_{i-1} forward measure, i.e.,

$$\mathbb{E}^{T_{i-1}}[L(T_{i-1}, T_i)] \neq \mathbb{E}^{T_i}[L(T_{i-1}, T_i)]$$

the difference between these two expectations is commonly referred to as **convexity**.

Convexity Correction

Remember

$$\frac{d\mathbb{Q}^{T_i}}{d\mathbb{Q}^{T_{i-1}}}\Big|_{\mathcal{F}(t_{i-1})} = \frac{P(T_{i-1}, T_i)}{P(t_0, T_i)} \frac{P(t_0, T_{i-1})}{P(T_{i-1}, T_{i-1})}$$

By the change of measure technique, we can simplify the equation [3] such that:

$$\begin{aligned} V(t_0) &= NP(t_0, T_{i-1})\mathbb{E}^{T_{i-1}}[L(T_{i-1}, T_i)] \\ &= NP(t_0, T_{i-1})\mathbb{E}^{T_i}\left[L(T_{i-1}, T_i)\frac{P(t_0, T_i)}{P(T_{i-1}, T_i)}\frac{P(T_{i-1}, T_{i-1})}{P(t_0, T_{i-1})}\right] \\ &= N\mathbb{E}^{T_i}\left[L(T_{i-1}, T_i)\frac{P(t_0, T_i)}{P(T_{i-1}, T_i)}\right] \end{aligned}$$

Convexity Correction

then by simply adding and subtracting $L(T_{i-1}, T_i)$ we get:

$$V(t_0) = N\mathbb{E}^{T_i}[L(T_{i-1}, T_i)] + N\mathbb{E}^{T_i}\left[L(T_{i-1}, T_i)\left(\frac{P(t_0, T_i)}{P(T_{i-1}, T_i)} - 1\right)\right]$$

with the second term being the convexity correction between the two maturities

$$\begin{aligned} cc(T_{i-1}, T_i) &= \mathbb{E}^{T_i}\left[L(T_{i-1}, T_i)\left(\frac{P(t_0, T_i)}{P(T_{i-1}, T_i)} - 1\right)\right] \\ &= P(t_0, T_i)\mathbb{E}^{T_i}\left[\frac{L(T_{i-1}, T_i)}{P(T_{i-1}, T_i)}\right] - F(t_0; T_{i-1}, T_i) \end{aligned}$$

Convexity Correction

recall the definition of the simple Libor rate $L(T_{i-1}, T_i)$:

$$P(T_{i-1}, T_i) = \frac{1}{1 + \tau_i L(T_{i-1}, T_i)}$$

then the expectation inside the $cc(T_{i-1}, T_i)$ equation can be written as follows

$$\begin{aligned}\mathbb{E}^{T_i}\left[\frac{L(T_{i-1}, T_i)}{P(T_{i-1}, T_i)}\right] &= \mathbb{E}^{T_i}\left[L(T_{i-1}, T_i) + \tau_i L^2(T_{i-1}, T_i)\right] \\ &= F(t_0, T_{i-1}, T_i) + \tau_i \mathbb{E}^{T_i}[L^2(T_{i-1}, T_i)]\end{aligned}$$

Convexity Correction

Exercise:

We know that $L(T_{i-1}, T_i)$ is a martingale under the T_i forward measure, is the quantity $L^2(T_{i-1}, T_i)$ a martingale too under the same measure?

Hint:

Consider the dynamics

$$dL_i(t) = \sigma L_i(t) dW_i^i(t)$$

and apply Ito's Lemma to $L_i^2(t)$, what can you say about the drift of the SDE?